DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE

SVKM's Dwarkadas J. Sanghvi College of Engineering (DJSCE), Mumbai

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TUTORIAL 1: LITERATURE REVIEW AND RESEARCH GAP

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Project Title:	FinReg-SLM: Domain-Specialized Small Language Model for Multi-Agency Indian Financial Regulatory Compliance
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PART B — Project Title, Background & Problem Statement

1. Project Title

Formal Title: FinReg-SLM: A Version-Aware, Multi-Agency Retrieval-Augmented Fine-Tuned Small Language Model with Native Chain-of-Thought Reasoning and Direct Preference Optimization for Indian Financial Regulatory Compliance

2. Background and Motivation

The Multi-Agency Compliance Landscape: Indian financial institutions operate within a dense, multi-jurisdictional statutory ecosystem governed by 13 primary regulatory authorities (including RBI, SEBI, MCA, CBIC, CBDT, IBBI, EPFO, ESIC, IRDAI, PFRDA, MeitY, FIU-IND, and IFSCA) alongside 25 governing Parliamentary Acts (e.g., Banking Regulation Act 1949, RBI Act 1934, Companies Act 2013, SEBI Act 1992, DPDP Act 2023). Together, over 18,242 active statutory circulars, Master Directions, notifications, and gazette orders impose rigorous operational mandates on scheduled commercial banks, NBFCs, and fintech enterprises.

Temporal Volatility and Superseded Clauses: Regulatory circulars are subject to continuous partial amendments, clarifications, and sunset clauses. For instance, a 2025 SEBI amendment to the Takeover Regulations modified share valuation guidelines while granting a 9-month transitional window. Standard semantic vector search fails in this setting: older 2018 circulars often exhibit higher cosine similarity to user queries than the recent amendment, causing standard RAG systems to retrieve obsolete law and generate hallucinated compliance advice.

> **📌 Data Sovereignty & Legal Mandate (DPDP Act 2023)**
> Deploying offshore cloud-hosted LLM APIs (such as OpenAI GPT-4o or Google Gemini 2.5) introduces severe statutory liability. Uploading internal financial audit trails, banking logs, and customer KYC records to third-party offshore endpoints violates data localization and privacy mandates under India's Digital Personal Data Protection (DPDP) Act 2023. Financial enterprises require a 100% on-premises, self-hosted Small Language Model (SLM) executing on local institutional hardware.

Epistemic Limits and Hallucination Risks: General-purpose language models suffer from three acute failure modes when querying statutory circulars: (1) Hallucination of Authority, where models invent plausible circular reference numbers; (2) Distractor Sensitivity, where irrelevant text in the retrieval window misleads output generation; and (3) Epistemic Sycophancy, where models answer questions based on false premises rather than rejecting them.

The FinReg-SLM Approach: FinReg-SLM addresses these fundamental challenges by adapting DeepSeek-R1-Distill-Qwen-7B for local execution inside 4.5 GB VRAM on a consumer 8GB GPU. The system combines a version-aware hybrid retrieval engine (Dense BGE-small + BM25 FTS5 + Reciprocal Rank Fusion k=60 + Temporal Filter) with a Two-Stage Alignment strategy (Stage 1 QLoRA SFT on Chain-of-Thought traces + Stage 2 Native DPO on epistemic preference pairs).

3. Problem Statement

Standardized Statement: Indian financial institutions operating under 13 statutory bodies lack automated, version-aware compliance verification tools. Existing general-purpose LLMs exhibit a 17%–33% hallucination rate, fail to track statutory circular amendments, confabulate answers on out-of-scope inquiries, and violate data sovereignty mandates under the DPDP Act 2023. This project designs, implements, and benchmarks an on-premises Small Language Model architecture providing verifiable citation grounding, calibrated epistemic abstention, and version-aware multi-agency reasoning on consumer-grade hardware.

PART C — Literature Search Methodology

Primary Boolean Search Query: ('Retrieval-Augmented Generation' OR 'RAFT') AND ('Fine-Tuning' OR 'LoRA' OR 'DPO') AND ('Legal NLP' OR 'Financial Compliance' OR 'Regulatory QA') AND ('Hallucination' OR 'Catastrophic Forgetting' OR 'Temporal Validity')

Databases and Venues Searched: Scopus, ScienceDirect (Elsevier), IEEE Xplore, ACM Digital Library, and arXiv (for latest benchmarks). Inclusion criteria prioritized peer-reviewed articles and top AI conferences (EMNLP, NeurIPS, COLM, IEEE Access, Elsevier Q1) published between 2023 and 2026 with reproducible quantitative metrics.

PART D — Core Literature Review (Top 8 Papers)

7. Table 1: Index of Primary Literature Sources

Sr.	Authors & Year	Paper Title	Venue	Year	Indexing	DOI / Link
1	**Ovadia et al. [1]**	Fine-Tuning or Retrieval? Comparing Knowledge Injection in LLMs	EMNLP 2024	2024	Scopus / WoS	10.18653/v1/2024.emnlp-main.15
2	**Zhang et al. [2]**	RAFT: Adapting Language Model to Domain Specific RAG	COLM 2024	2024	Core AI Conf.	openreview:rzQGHXNReU
3	**Magesh et al. [3]**	Hallucination-Free? Assessing Reliability of AI Legal Research Tools	J. Empir. Leg. Stud.	2025	Scopus Q1	10.1111/jels.12413
4	**Pai & Hsu [4]**	Fine-Tuning SLMs for Industrial RAG: Factuality & Hallucination Analysis	Comput. Stand. Interfaces	2026	Scopus Q1	10.1016/j.csi.2026.104163
5	**Kim [5]**	Detecting Jurisdiction Compliance Gaps with Regulatory RAG	J. Loss Prev. Proc. Ind.	2026	Scopus Q1	10.1016/j.jlp.2026.106100
6	**Rafailov et al. [6]**	Direct Preference Optimization: Language Model Is Secretly a Reward Model	NeurIPS 2023	2023	Scopus / NeurIPS	neurips:a85b405

7	**Biderman et al. [7]**	LoRA Learns Less and Forgets Less	TMLR	2024	Scopus / TMLR	openreview:c806jB5m5K
8	**Pall [8]**	IndiaFinBench: Benchmark for LLMs on Indian FinReg Text	arXiv:2604.19298	2026	Benchmark Dataset	arXiv:2604.19298

8. Table 2: 10-Column Comparative Literature Matrix (Problem, Gap, Fix & Impact)

Sr.	Authors	Research Problem	Dataset	Method	Key Findings	Research Gap	Our Applied Fix	Impact on Score	Benchmark Gain
1	**Ovadia et al. [1]**	RAG vs Fine-Tuning for knowledge injection	Synthetic point-facts & QA	Empirical SFT vs RAG comparison	RAG beats SFT on factuality; FT suffers from Reversal Curse	FT alone fails for rapidly amending regulations	Two-Stage RAFT with <think> traces + Distractor Ingestion	Forces reasoning over retrieved context rather than memorization	**Core Compliance: 62.3% -> 94.43%**
2	**Zhang et al. [2]**	Model confusion from irrelevant retrieved context	PubMed, HotpotQA, Gorilla	RAFT SFT with oracle + distractor docs	Distractor training forces verbatim citation extraction	General domain only; not tested on circulars	Extended RAFT to Indian multi-agency financial law	Eliminated distraction-induced hallucinations	**C2 Multi-hop: +40.44% gain**
3	**Magesh et al. [3]**	Hallucination in commercial legal AI tools	200+ real-world legal queries	Empirical pre-registered audit (Lexis, Westlaw)	17%–33% hallucination rate on judicial authority	Audited closed commercial tools without open SLM fix	Verbatim Citation Verification + DPO Margin Loss Penalty	Zero tolerance for fabricated circular reference numbers	**Citation Grounding: 51.6% -> 99.17%**
4	**Pai & Hsu [4]**	SLM hallucination on internal corporate norms	Industrial RAG datasets	QLoRA 1.5B–8B SLM fine-tuning	Domain-specific legal RAG is harder than customer service	Corporate norms only; no multi-agency statutory tracking	4-bit QLoRA on 6 failure classes across 13 regulators	Enables 7B SLM to execute on-premises in 4.5GB VRAM	**Cross-Reg (C3): 96.67% accuracy**
5	**Kim [5]**	Jurisdiction-specific compliance gaps	1,327 Safety Norms	SQLite FTS5 BM25 + Qwen-4B SLM	On-premises SLM raised gap detection 0/60 -> 26/60	Single-agency domain; no temporal version tracking	SQLite WAL index + explicit effective_date metadata filter	Prevents retrieving superseded older circulars	**Contradiction Detection: 79.00%**
6	**Rafailov et al. [6]**	RL instability in LLM preference alignment	HH-RLHF, Anthropic data	Direct Preference Optimization (DPO)	DPO optimizes policy directly without separate reward model	General preference; no legal refusal/sycophancy tuning	257 Domain DPO Pairs with Negative Traps (C5/C6)	Forces model to actively refuse when statutory context lacks facts	**Refusal (C5): 0.0% -> 82.50%**
7	**Biderman et al. [7]**	Catastrophic forgetting under fine-tuning	1B–7B open LLMs	LoRA forgetting audit across NLP benchmarks	Full FT degrades general reasoning; LoRA preserves base skills	Did not test LoRA rank on all 7 linear projection matrices	All-Linear QLoRA (r=64, a=128) in 4-bit NF4	Preserves base DeepSeek-R1 reasoning and syntax formatting	**Protocol Conformance: 100.0%**
8	**Pall [8]**	Benchmarking LLMs on Indian financial regulatory text	192 SEBI & RBI circulars	Zero-shot benchmark (324 Questions)	Base 7B/8B models score ~76%; numerical reasoning is hardest	Released benchmark only; no fine-tuned local SLM proposed	Evaluated Stage 2 DPO model zero-shot on 324 questions	Outperformed Base LLaMA-3-8B and Mistral-7B locally	**Overall Macro: 80.98% (Top Tier 2)**

PART E — Literature Synthesis

1. SLMs vs. Cloud Models: Domain-adapted Small Language Models (3B–8B) using Parameter-Efficient Fine-Tuning (QLoRA) and Direct Preference Optimization achieve specialized statutory accuracy while maintaining on-premises data privacy and zero API inference costs.

2. Hybrid Retrieval & Temporal Validity: Lexical BM25 and dense semantic embeddings fused via Reciprocal Rank Fusion (k=60) coupled with explicit date filtering (effective_date) eliminate temporal obsolescence in dynamic regulatory circulars.

3. Epistemic Abstention & Sycophancy: Training models exclusively on positive QA pairs induces severe sycophancy. Incorporating negative preference pairs forces the model to actively refuse when statutory authority is absent.

4. Data Sovereignty under DPDP Act 2023: Deploying 4-bit quantized 7B models on local hardware eliminates third-party cloud data leakage, meeting strict regulatory compliance mandates under the DPDP Act 2023.

PART F — Research Gaps, Applied Fixes & Empirical Gains

• **Gap 1 (RAG vs FT Dichotomy [1, 2]):** Naive RAG is confused by legal distractors; pure fine-tuning suffers from factual hallucinations. **Our Applied Fix: Two-Stage RAFT with <think> traces (Golden Chunks + Distractor Circulars). Empirical Gain: Core statutory compliance rose to 94.43%; IndiaFinBench Regulatory Interpretation scored 90.29%.**

• **Gap 2 (Hallucination of Authority [3, 4]):** Commercial AI hallucinates legal circular numbers in 17%–33% of queries. **Our Applied Fix: Verbatim Citation Verification + DPO Margin Loss Penalty. Empirical Gain: Citation grounding reached 99.17%; phantom citations dropped to 0.0%.**

• **Gap 3 (Temporal Amnesia in RAG [5]):** Standard vector search retrieves obsolete, superseded circulars based on semantic similarity. **Our Applied Fix: Date metadata filtering + RRF (k=60) + temporal DPO. Empirical Gain: Contradiction Detection scored 79.00%; Temporal Reasoning scored 75.78%.**

• **Gap 4 (Epistemic Inability to Abstain [3, 6]):** Models answer ungrounded or false-premise questions sycophantically. **Our Applied Fix: Negative preference DPO alignment (C5 Refusal & C6 False Premise Pairs). Empirical Gain: Refusal quality rose from 0.0% to 82.50%; premise rejection to 58.00%.**

• **Gap 5 (Catastrophic Forgetting [7]):** Fine-tuning degrades reasoning and formatting. **Our Applied Fix: All-linear QLoRA (r=64, a=128) in 4-bit NF4 with strict ChatML Token Supervision. Empirical Gain: 100.0% structural ChatML protocol conformance; zero syntax collapse.**

• **Gap 6 (Non-Western Regulatory Evaluation Void [8]):** NLP benchmarks ignore Indian financial law. **Our Applied Fix: Dual Evaluation on internal G7 Suite (300 Qs) and external IndiaFinBench (324 Qs). Empirical Gain: 80.98% macro accuracy, outperforming open base 7B/8B models by +3.4% to +5.0%.**

PART G & H — Research Question & Objectives

> **Central Research Question**
> How does combining a version-aware multi-agency hybrid retrieval pipeline with a Two-Stage Aligned Reasoning Model (Stage 1 QLoRA SFT on Chain-of-Thought traces + Stage 2 Native DPO on epistemic preference pairs using DeepSeek-R1-Distill-Qwen-7B) compare against baseline small language models in achieving factual accuracy, 100% citation grounding, and calibrated epistemic abstention across 13 Indian financial regulatory authorities under on-premises DPDP Act 2023 constraints?

• **1. Ingestion Pipeline:** Index 18,242 verified documents across 13 Indian authorities in SQLite WAL mode.

• **2. Hybrid Retrieval Engine:** Implement Dense (BGE-small) + BM25 FTS5 with RRF (k=60) and temporal date filtering.

• **3. Dataset Synthesis:** Synthesize 257 golden CoT reasoning traces and 257 DPO preference pairs across 6 failure classes.

• **4. Two-Stage Fine-Tuning:** Adapt DeepSeek-R1-7B on an 8GB consumer GPU using 4-bit NF4 QLoRA and native DPO.

• **5. Local Deployment Engine:** Export standalone FP16 weights (15.2 GB) and build an on-premises FastAPI backend with streaming SSE.

• **6. Empirical Evaluation:** Benchmark on 300 G7 queries and 324 IndiaFinBench questions via a 70B LLM Judge.

PART I & J — Scope & Empirical Benchmark Results

Table A: Internal G7 Hard Suite Benchmark (300 Queries via LLaMA-3.3-70B Judge)

Model Configuration	C1	C2	C3	C4	C5 (Refuse)	C6 (Reject)	Citation	Judge Score
Base 3B (Llama-3.2-3B)	58.0%	52.0%	46.0%	44.0%	0.0%	0.0%	18.6%	44.80%
RAFT Fine-Tuned 3B	72.0%	68.0%	64.0%	60.0%	0.0%	0.0%	78.0%	18.80%*
Stage 1 SFT 7B (DeepSeek-R1)	90.0%	92.0%	89.5%	91.0%	0.0%	0.0%	88.4%	68.17%
Stage 2 DPO 7B (Our Final Model)	**92.2%**	**94.4%**	**96.7%**	**93.8%***	**82.5%**	**58.0%**	**99.2%**	**70.71%**

Table B: Complete IndiaFinBench Test Split Leaderboard (arXiv:2604.19298 — 324 Questions)

Model Name	Hosting / Size	Regulatory	Contradiction	Temporal	Numerical	Overall	Standing Tier
Gemini 2.5 Flash	Cloud MoE (Proprietary)	93.40%	88.00%	86.50%	87.20%	89.70%	Tier 1 (Cloud API)
GPT-4o	Cloud MoE (Proprietary)	92.10%	86.00%	84.40%	85.10%	88.20%	Tier 1 (Cloud API)
Claude 3.5 Sonnet	Cloud MoE (Proprietary)	91.00%	84.00%	82.80%	83.50%	86.50%	Tier 1 (Cloud API)
FinReg-SLM 7B (Ours)	**7B Dense (Local 8GB)**	**90.29%**	**79.00%**	**75.78%**	**69.18%**	**80.98%**	**Top Tier 2 (Ours)**
LLaMA-3-8B Base	8B Dense (Open Base)	84.20%	72.50%	70.10%	58.40%	77.60%	Tier 2 (Open Base)
Mistral-7B-Instruct	7B Dense (Open Base)	82.80%	71.00%	68.50%	56.20%	76.20%	Tier 2 (Open Base)
Gemma 4 E4B	4B Dense (Compact)	76.50%	65.00%	62.00%	51.00%	70.40%	Tier 3 (Compact)
Non-Specialist Human	Human Baseline	74.00%	66.00%	65.00%	54.00%	69.00%	Human Baseline

15. Future Scope & Institutional Scaling Roadmap

• **Scaling to 14B/32B/70B Open Architectures:** Fine-tuning DeepSeek-R1-Distill-14B/32B or LLaMA-3.3-70B on institutional multi-GPU compute (A100/H100) will expand working memory, projected to achieve >91.5% accuracy and surpass proprietary cloud models on-premises.

• **Program-Aided Compliance (Python REPL Sandbox):** Integrating an automated Python execution sandbox inside <think> CoT traces to execute multi-step calculations, pushing numerical reasoning from 69.18% to 98%+.

• **Online Group Relative Policy Optimization (GRPO):** Transitioning to online GRPO with rule-based reward functions for statutory circular citations and mathematical verification.

• **Autonomous 24/7 Regulatory Monitoring Daemon:** Deploying background daemons to monitor daily gazettes, updating the SQLite WAL index with real-time SSE compliance alerts.

PART K — References (IEEE Format)

[1] O. Ovadia, M. Brief, M. Mishaeli, and O. Elisha, 'Fine-Tuning or Retrieval? Comparing Knowledge Injection in LLMs,' in Proc. EMNLP 2024, pp. 237–250, 2024, doi: 10.18653/v1/2024.emnlp-main.15.

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[5] Y. Kim, 'Detecting jurisdiction compliance gaps in process plant engineering with regulatory RAG,' Journal of Loss Prevention in the Process Industries, vol. 103, p. 106100, 2026, doi: 10.1016/j.jlp.2026.106100.

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